A Descriptive Analysis

This section provides descriptive statistics for the numeric and categorical variables present in the Marilu City sales database used for model development. It includes measures of central tendency (Mean, Median), dispersion (Min, Max), and distribution (1st Qu., 3rd Qu.) for numerical variable and frequency tables for categorical variables.

A.1 Summary Table for Numerical Variable

Table A1: Summary Statistics for Key Numeric Variables

Statistic	TASP	landval	imprval	totalval	mbyrblt	flrarea1
Min.	68,077	24,751	25,817	52,463	1937	552
1st Qu.	125,224	46,363	75,056	124,756	1984	960
Median	144,440	52,350	88,808	142,922	1994	1,058
Mean	148,954	52,065	92,502	144,567	1991	1,068
3rd Qu.	167,858	57,045	107,139	161,884	1998	1,152
Max.	305,480	78,785	237,134	299,767	2015	1,978

TASP = Time-Adjusted Sale Price

Table A2: Summary Statistics for Key Numeric Variables (Part 2)

Statistic	lot_size	effage	bedrooms	baths	fireplcs	gar_area	bsmtarea
Min.	3,300	2	2	1	0	0	0
1st Qu.	6,123	19	3	1	0	0	951
Median	6,839	23	3	1.5	1	0	1,040
Mean	7,194	26	3	1.6	1	90	999
3rd Qu.	7,800	33	3	1.75	1	0	1,140
Max.	15,650	80	4	2.75	2	756	1,978

A.2 Frequency Table for Categorical Variable

Table A3: Frequency Distribution: Neighbourhood (neigh)

Neighbourhood	Count (n)	Percentage
716	156	33.5%
713	92	19.7%
737	83	17.8%
715	59	12.7%
731	29	6.2%
736	27	5.8%
734	12	2.6%
732	8	1.7%
Total	466	100.0%

Table A4: Frequency Distribution: Heat Type (heattype)

Heat Type Code	Count (n)	Percentage
2	403	86.5%
3	14	3.0%
4	12	2.6%
6	12	2.6%
8	8	1.7%
5	7	1.5%
7	7	1.5%
1	3	0.6%
Total	466	100.0%

Table A5: Frequency Distribution: Garage Type (gar_type)

Garage Type Code	Count (n)	Percentage
0	333	71.5%
2	57	12.2%
1	43	9.2%
3	33	7.1%
Total	466	100.0%

Table A6: Frequency Distributions for Binary Site/Improvement Features (Part 1)

Variable	Level	Count (n)	Percentage
baseste (Basement)	0 (No)	442	94.8%
	1 (Yes)	24	5.2%
garagefl (Garage)	0 (No)	333	71.5%
	1 (Yes)	133	28.5%
carptfl (Carport)	0 (No)	268	57.5%
	1 (Yes)	198	42.5%
conduit (Wiring)	0 (Overhead)	166	35.6%
	1 (Underground)	300	64.4%
alley	0 (No)	318	68.2%
	1 (Yes)	148	31.8%

Table A7: Frequency Distributions for Binary Site/Improvement Features (Part 2)

Variable	Level	Count (n)	Percentage
sidewalk	0 (No)	423	90.8%
	1 (Yes)	43	9.2%
culdesac	0 (No)	437	93.8%
	1 (Yes)	29	6.2%
corner	0 (No)	394	84.5%
	1 (Yes)	72	15.5%
wooded	0 (No)	430	92.3%
	1 (Yes)	36	7.7%
pie_shpe (Pie Shape)	0 (No)	439	94.2%
	1 (Yes)	27	5.8%

Table A8: Frequency Distributions for Binary Site/Improvement Features (Part 3)

Variable	Level	Count (n)	Percentage
slope	0 (Level)	406	87.1%
	1 (Sloped)	60	12.9%
abv_road (Above Road)	0 (No/Level)	385	82.6%
	1 (Yes)	81	17.4%
splitlvl (Split Level)	0 (No)	432	92.7%
	1 (Yes)	34	7.3%

B Time Adjustment

Table A9: Kruskal-Wallis Test Statistics: Time-Adjusted Sale Price (TASP) by Sale Month

Statistic	Value
Kruskal-Wallis chi-squared	9.970
df	7
p-value	0.190

Table A10: Correlation Matrix for Binary Site Variables

Variable	conduit	alley	sidewalk	culdesac	corner	wooded	pie_shpe	slope	abv_road
conduit	1.000	-0.811	−0.289	0.192	−0.017	0.064	0.165	0.005	0.010
alley	-0.811	1.000	0.228	−0.157	−0.049	−0.077	−0.149	−0.056	−0.106
sidewalk	−0.289	0.228	1.000	−0.082	0.007	−0.092	−0.047	−0.078	0.030
culdesac	0.192	−0.157	−0.082	1.000	−0.086	0.059	0.430	0.007	−0.001
corner	−0.017	−0.049	0.007	−0.086	1.000	−0.035	−0.030	−0.076	−0.071
wooded	0.064	−0.077	−0.092	0.059	−0.035	1.000	0.031	−0.015	−0.069
pie_shpe	0.165	−0.149	−0.047	0.430	−0.030	0.031	1.000	−0.095	−0.090
slope	0.005	−0.056	−0.078	0.007	−0.076	−0.015	−0.095	1.000	0.669
abv_road	0.010	−0.106	0.030	−0.001	−0.071	−0.069	−0.090	0.669	1.000

Note: Values below -0.8 are highlighted in red.

C Land Value Estimation

Table A11: Ratio Statistics for Land Ratio by Conduit (After Adjustment)

Conduit	Median	Mean	LB Mean	UB Mean	Min	Max	COD (%)	CV (%)	LB Median	UB Media
0	1.00	1.01	0.98	1.05	0.41	1.75	19.10	24.20	0.94	1.05
1	1.00	0.99	0.96	1.01	0.12	1.60	18.60	24.00	0.98	1.03

Table A12: Kruskal-Wallis Test: Adjusted Land Ratio ($LandRatio \times ConduitFactor$) by Conduit (After Adjustment)

Statistic	Value
Kruskal-Wallis chi-squared	0.493
df	1
p-value	0.483

Table A13: Kruskal-Wallis Test: Fully Adjusted Land Ratio by Neighbourhood (After Adjustment)

Statistic	Value
Kruskal-Wallis chi-squared	4.891
df	7
p-value	0.673

D Cost Model

D.1 Depreciation Analysis

Table A14: Model Summary: Quadratic Depreciation Model (Dependent: obsdeprn)

	Model Summary					
Equation	R Square	Adj. R Sq.	F	df1	df2	Sig. (F-test)
Quadratic	0.207	0.203	59.740	2.000	459.000	<0.001

Dependent Variable: obsdeprn (filtered for positive values)

Independent Variable: effage2, I(effage2^2)

Table A15: Model Summary: Cubic Depreciation Model (Dependent: obsdeprn)

	Model Summary					
Equation	R Square	Adj. R Sq.	F	df1	df2	Sig. (F-test)
Cubic	0.209	0.204	40.390	3.000	458.000	<0.001

Dependent Variable: obsdeprn (filtered for positive values)

Independent Variable: effage2, I(effage2^2), I(effage2^3)

Table A16: Cubic Depreciation Model Coefficients (Dependent: obsdeprn)

Term	Estimate	Std. Error	t value	Sig. (p-value)
(Intercept)	1.26×10^{-1}	4.50×10^{-2}	2.809	0.005
effage2	6.37×10^{-3}	6.83×10^{-3}	0.933	0.351
I(effage2^2)	2.42×10^{-4}	3.31×10^{-4}	0.730	0.466
I(effage2^3)	-6.03×10^{-6}	4.85×10^{-6}	-1.244	0.214

D.2 Market Factor Adjustment

Table A17: Contingency Table: Garage Floor (garagefl) vs Garage Type (gar_type)

	Garage Type (gar_type)				
Garage Floor (garagefl)	0 (None)	1 (Att.)	2 (Det.)	3 (Bsmt)	Row Total
0 (No)	333	0	0	0	333
1 (Yes)	0	43	57	33	133
Column Total	333	43	57	33	466

Table A18: Chi-Squared Test: Garage Floor (garagefl) vs Garage Type (gar_type)

Statistic	Value
Pearson Chi-squared	466.000
df	3
p-value	< 2.2e-16

Note: Test confirms strong association.

Warning: Chi-squared approximation may be inaccurate due to zero cells.

Table A19: Ratio Statistics for Market Factor (Improvement Ratio) by Binary Variables (Before Adjustment)

Variable	Level	Median	COD (%)	LB Median	UB Median
conduit	0	0.98	15.40	0.95	1.00
	1	1.00	10.60	0.98	1.02
alley	0	1.00	10.90	0.98	1.01
	1	0.99	15.40	0.96	1.00
sidewalk	0	0.99	12.30	0.98	1.01
	1	0.99	13.50	0.96	1.07
culdesac	0	0.99	12.30	0.98	1.00
	1	1.05	12.10	0.95	1.13
corner	0	0.99	12.30	0.97	1.00
	1	1.02	12.60	0.96	1.06
wooded	0	0.99	12.60	0.98	1.01
	1	1.00	9.99	0.94	1.04
pie_shpe	0	0.99	12.40	0.98	1.00
	1	1.05	11.50	0.95	1.12
slope	0	0.99	12.50	0.97	1.00
	1	1.01	11.70	0.95	1.03
abv_road	0	0.99	12.60	0.97	1.01
	1	1.00	11.40	0.96	1.02

Table A20: Ratio Statistics for Market Factor by Garage Type (After Adjustment)

gar_type	Median	Mean	LB Mean	UB Mean	Min	Max	COD (%)	CV (%)	LB Median	UB Median
0 (None)	1.00	1.00	0.99	1.02	0.48	1.70	12.23	16.42	0.99	1.01
1 (Attached)	0.97	1.00	0.96	1.04	0.78	1.25	11.03	13.22	0.95	1.05
2 (Detached)	1.00	0.98	0.93	1.02	0.66	1.47	13.06	17.54	0.94	1.02
3 (Basement)	1.00	0.99	0.95	1.04	0.78	1.22	9.61	11.95	0.94	1.03

Table A21: Kruskal-Wallis Test: Adjusted Market Factor by Garage Type (After Adjustment)

Statistic	Value
Kruskal-Wallis chi-squared	0.828
df	3
p-value	0.849

Table A22: Ratio Statistics for Garage-Adjusted Market Factor by Heat Type (Before Merge)

Heat Type	Median	Mean	LB Mean	UB Mean	Min	Max	COD (%)	CV (%)	LB Median	UB Median
1	1.14	1.14	0.97	1.32	1.08	1.22	4.03	6.03	NA	NA
2	0.99	1.00	0.98	1.01	0.48	1.70	12.0	15.9	0.98	1.01
3	1.11	1.17	1.05	1.28	0.84	1.56	15.2	17.4	1.01	1.33
4	1.00	1.00	0.89	1.10	0.78	1.30	12.2	16.0	0.83	1.04
5	0.97	0.95	0.81	1.09	0.63	1.12	9.55	16.3	0.63	1.05
6	1.01	1.01	0.94	1.08	0.81	1.20	8.26	11.1	0.93	1.05
7	0.94	0.95	0.84	1.06	0.78	1.14	9.35	12.6	0.78	1.04
8	0.96	0.95	0.84	1.07	0.77	1.14	11.2	14.2	0.77	1.12

Table A23: Ratio Statistics for Garage-Adjusted Market Factor by Heat Type (After Merging Type 1 into Type 2)

Heat Type	Median	Mean	LB Mean	UB Mean	Min	Max	COD (%)	CV (%)	LB Median	UB Median
2 (Merged 1&2)	0.99	1.00	0.98	1.01	0.48	1.70	12.06	15.88	0.98	1.01
3	1.11	1.17	1.05	1.28	0.84	1.56	15.18	17.36	1.01	1.33
4	1.00	1.00	0.89	1.10	0.78	1.30	12.18	16.05	0.83	1.04
5	0.97	0.95	0.81	1.09	0.63	1.12	9.55	16.28	0.63	1.05
6	1.01	1.01	0.94	1.08	0.81	1.20	8.26	11.09	0.93	1.05
7	0.94	0.95	0.84	1.06	0.78	1.14	9.35	12.61	0.78	1.04
8	0.96	0.95	0.84	1.07	0.77	1.14	11.20	14.17	0.77	1.12

Table A24: Kruskal-Wallis Test Results: Fully Adjusted Market Factor by Binary Variables (After Garage & Heat Type Adjustments)

Variable Tested	p-value
heattype	0.548
baseste	0.153
garagefl	0.300
carptfl	0.322
gar_type	0.523
splitlvl	0.124

Table A25: Ratio Statistics for Market Factor by Heat Type (After Garage & Heat Type Adjustments)

Heat Type	Median	Mean	LB Mean	UB Mean	Min	Max	COD (%)	CV (%)	LB Median	UB Median
2 (Merged 1&2)	0.99	1.00	0.98	1.01	0.48	1.70	12.06	15.88	0.98	1.01
3	1.00	1.05	0.95	1.16	0.76	1.41	15.18	17.36	0.91	1.20
4	1.00	1.00	0.89	1.10	0.78	1.30	12.18	16.05	0.83	1.04
5	0.97	0.95	0.81	1.09	0.63	1.12	9.55	16.28	0.63	1.05
6	1.01	1.01	0.94	1.08	0.81	1.20	8.26	11.09	0.93	1.05
7	1.00	1.01	0.90	1.13	0.83	1.21	9.35	12.61	0.83	1.11
8	0.96	0.95	0.84	1.07	0.77	1.14	11.20	14.17	0.77	1.12

Table A26: Correlation Matrix for Key Non-Binary Improvement Variables

Variable	baseste	carptfl	bedrooms	familyrm	fireplcs	cp_mancl	gr_mancl
baseste	1.000	−0.004	−0.006	−0.135	−0.043	−0.004	0.089
carptfl	−0.004	1.000	0.041	0.089	0.237	1.000	−0.380
bedrooms	−0.006	0.041	1.000	0.153	0.180	0.041	0.097
familyrm	−0.135	0.089	0.153	1.000	0.402	0.089	0.119
fireplcs	−0.043	0.237	0.180	0.402	1.000	0.238	0.145
cp_mancl	−0.004	1.000	0.041	0.089	0.238	1.000	−0.380
gr_mancl	0.089	−0.380	0.097	0.119	0.145	−0.380	1.000

Note: Values equal above 0.7 are highlighted in red.

Table A27: Kruskal-Wallis Test: Adjusted Improvement Ratio by Stories

Statistic	Value
Kruskal-Wallis chi-squared	2.056
df	1
p-value	0.152

Table A28: Ratio Statistics: Adjusted Improvement Ratio by Stories

Stories	Median	Mean	LB Mean	UB Mean	Min	Max	COD (%)	CV (%)	LB Median	UB Median
1	0.99	0.99	0.98	1.01	0.48	1.70	11.68	15.36	0.98	1.00
2	1.01	1.05	0.98	1.13	0.68	1.52	16.13	19.38	0.95	1.16

Table A29: Frequency Distribution: Original Main Building Manual Class (mbmancls)

Class	40	51	80	81	90	92	140	142	145	147	150
Count	8	3	38	9	222	5	158	10	4	3	6

Table A30: Kruskal-Wallis Test: Adjusted Improvement Ratio by Grouped mbmancls (Before Adjustment)

Statistic	Value
Kruskal-Wallis chi-squared	15.307
df	3
p-value	0.0017

Table A31: Ratio Statistics: Adjusted Improvement Ratio by Grouped mbmancls (Before Adjustment)

Group	Median	Mean	LB Mean	UB Mean	Min	Max	COD (%)	CV (%)	LB Median	UB Median
14	1.00	1.01	0.99	1.03	0.63	1.39	10.15	12.57	0.98	1.02
15	0.95	0.95	0.88	1.03	0.78	1.21	11.44	13.52	0.83	1.03
8	1.08	1.08	1.01	1.14	0.58	1.70	17.67	22.15	1.01	1.16
9	0.98	0.97	0.95	0.99	0.48	1.39	11.22	14.96	0.96	1.00

Table A32: Kruskal-Wallis Test: Adjusted Improvement Ratio by Grouped mbmancls (After Adjustment)

Statistic	Value
Kruskal-Wallis chi-squared	6.857
df	3
p-value	0.078

Table A33: Kruskal-Wallis Test: Adjusted Improvement Ratio by Family Room (Before Adjustment)

Statistic	Value
Kruskal-Wallis chi-squared	8.675
df	2
p-value	0.013

Table A34: Ratio Statistics: Adjusted Improvement Ratio by Family Room (Before Adjustment)

familyrm	Median	Mean	LB Mean	UB Mean	Min	Max	COD (%)	CV (%)	LB Median	UB Median
0	0.95	0.96	0.94	0.99	0.48	1.57	13.95	17.94	0.93	1.00
1	1.00	1.00	0.98	1.02	0.54	1.41	10.65	13.80	0.98	1.01
2	1.02	1.02	0.97	1.07	0.80	1.22	8.67	11.29	0.97	1.07

Table A35: Kruskal-Wallis Test: Adjusted Improvement Ratio by Family Room (After Adjustment)

Statistic	Value
Kruskal-Wallis chi-squared	0.806
df	2
p-value	0.668

Table A36: Ratio Statistics: Adjusted Improvement Ratio by Family Room (After Adjustment)

familyrm	Median	Mean	LB Mean	UB Mean	Min	Max	COD (%)	CV (%)	LB Median	UB Median
0	1.00	1.02	0.99	1.04	0.51	1.66	13.95	17.94	0.98	1.05
1	1.00	1.00	0.98	1.02	0.54	1.41	10.65	13.80	0.98	1.01
2	1.02	1.02	0.97	1.07	0.80	1.22	8.67	11.29	0.97	1.07

Table A37: Kruskal-Wallis Test: Adjusted Improvement Ratio by Bedrooms (Before Adjustment)

Statistic	Value
Kruskal-Wallis chi-squared	7.390
df	2
p-value	0.025

Table A38: Ratio Statistics: Adjusted Improvement Ratio by Bedrooms (Before Adjustment)

bedrooms	Median	Mean	LB Mean	UB Mean	Min	Max	COD (%)	CV (%)	LB Median	UB Median
2	0.98	0.97	0.94	1.00	0.54	1.66	12.70	16.99	0.93	1.00
3	1.01	1.02	1.00	1.03	0.51	1.47	10.93	14.13	0.99	1.02
4	0.97	0.99	0.89	1.10	0.68	1.42	17.96	22.36	0.82	1.09

Table A39: Kruskal-Wallis Test: Adjusted Improvement Ratio by Bedrooms (After Adjustment)

Statistic	Value
Kruskal-Wallis chi-squared	2.675
df	2
p-value	0.265

Table A40: Ratio Statistics: Adjusted Improvement Ratio by Bedrooms (After Adjustment)

bedrooms	Median	Mean	LB Mean	UB Mean	Min	Max	COD (%)	CV (%)	LB Median	UB Median
2	1.00	0.99	0.96	1.02	0.55	1.69	12.70	16.99	0.95	1.02
3	1.01	1.02	1.00	1.03	0.51	1.47	10.93	14.13	0.99	1.02
4	0.97	0.99	0.89	1.10	0.68	1.42	17.96	22.36	0.82	1.09

E Sales Comparison

E.1 Variable Transformation

Table A41: Main Building Manual Class Factors (‘mb_factor‘)

Class Code(s) (‘mbmancls‘)	Factor
40, 41, 42	0.507
50, 51, 52	0.767
80, 81, 82	0.783
90, 91, 92	0.945
140, 141, 142	1.000
145, 146, 147	1.220
150, 151, 152	1.190
Other / Default	1.000

Table A42: Garage/Carport Manual Class Factors (‘cp_factor‘/‘gr_factor‘)

Class Code(s) (‘cp_mancl‘/‘gr_mancl‘)	Factor
910/911	0.450
920/921	1.000
930/931	1.200
940/941	1.420
Other / Default	1.000

E.2 Model Calibration

The following tables provide detailed coefficient estimates and collinearity statistics (Variance Inflation Factor - VIF, Tolerance) for the four MRA model variations tested.

Table A43: Model 1 Coefficients and Collinearity Statistics

Variable	Estimate	Std. Error	t value	Pr(>|t|)	VIF	Tolerance
(Intercept)	−3918.951	6697.106	−0.585	0.559	—	—
‘adjusted_land_size‘	9.475	0.891	10.640	<0.001	2.456	0.407
‘dlin_flrarea1‘	73.032	4.776	15.291	<0.001	2.510	0.398
‘dlin_flrarea2‘	49.974	7.312	6.834	<0.001	1.361	0.735
‘dlin_bsmtfin‘	18.802	3.970	4.736	<0.001	1.915	0.522
‘dlin_gar_area_attached‘	52.917	8.875	5.962	<0.001	1.383	0.723
‘dlin_gar_area_detached‘	29.903	8.071	3.705	<0.001	1.181	0.847
‘dlin_gar_area_basement‘	17.453	3.390	5.148	<0.001	1.636	0.611
‘dlin_cp_area‘	28.600	6.755	4.234	<0.001	1.481	0.675
‘oil_heattype‘	7028.551	3194.351	2.200	0.028	1.063	0.941
‘electric_heattype‘	−2143.111	2714.418	−0.790	0.430	1.059	0.944
‘baths‘	6847.027	1984.868	3.450	<0.001	2.252	0.444
‘bedrooms‘	3313.895	1497.598	2.213	0.027	1.262	0.793
‘fireplcs‘	2821.726	1138.787	2.478	0.014	1.600	0.625
‘baseste‘1	−3297.011	3284.356	−1.004	0.316	1.220	0.820
‘crwlarea‘	−10.214	3.272	−3.121	0.002	1.250	0.800
‘conduit‘1	−762.099	2302.038	−0.331	0.741	2.814	0.355

Note: Coefficient values populated from Model 1 output. Tolerance = 1/VIF.

Table A44: Model 2 Coefficients and Collinearity Statistics

Variable	Estimate	Std. Error	t value	Pr(>|t|)	VIF	Tolerance
(Intercept)	−6209.637	6767.838	−0.918	0.359	—	—
‘adjusted_land_size‘	9.595	0.738	12.995	<0.001	1.691	0.591
‘dlin_flrarea1‘	73.674	4.798	15.354	<0.001	2.538	0.394
‘dlin_flrarea2‘	49.961	7.303	6.841	<0.001	1.360	0.735
‘dlin_bsmtfin‘	19.130	3.983	4.803	<0.001	1.930	0.518
‘dlin_gar_area_attached‘	53.149	8.855	6.002	<0.001	1.379	0.725
‘dlin_gar_area_detached‘	29.031	8.122	3.574	<0.001	1.198	0.835
‘dlin_gar_area_basement‘	17.362	3.379	5.138	<0.001	1.628	0.614
‘dlin_cp_area‘	28.942	6.753	4.285	<0.001	1.483	0.674
‘oil_heattype‘	6866.435	3173.702	2.164	0.031	1.051	0.952
‘electric_heattype‘	−2032.022	2705.873	−0.751	0.453	1.054	0.948
‘baths‘	6641.158	1992.112	3.334	<0.001	2.273	0.440
‘bedrooms‘	3341.309	1496.588	2.233	0.026	1.262	0.792
‘fireplcs‘	2902.725	1141.417	2.543	0.011	1.610	0.621
‘baseste‘1	−3555.653	3282.735	−1.083	0.279	1.221	0.819
‘crwlarea‘	−10.459	3.254	−3.215	0.001	1.238	0.808
‘alley‘1	1859.840	1959.961	0.949	0.343	1.931	0.518

Note: Coefficient values populated from Model 2 output. Tolerance = 1/VIF.

Table A45: Model 3 Coefficients and Collinearity Statistics

Variable	Estimate	Std. Error	t value	Pr(>|t|)	VIF	Tolerance
(Intercept)	−419.210	7669.203	−0.055	0.956	—	—
‘adjusted_land_size‘	9.428	0.892	10.569	<0.001	2.464	0.406
‘dlin_flrarea1‘	79.702	8.573	9.297	<0.001	8.085	0.124
‘dlin_flrarea2‘	47.783	7.678	6.223	<0.001	1.501	0.666
‘dlin_bsmtfin‘	19.156	3.988	4.803	<0.001	1.932	0.518
‘dlin_gar_area_attached‘	53.005	8.877	5.971	<0.001	1.383	0.723
‘dlin_gar_area_detached‘	30.495	8.097	3.766	<0.001	1.188	0.842
‘dlin_gar_area_basement‘	17.130	3.408	5.027	<0.001	1.653	0.605
‘dlin_cp_area‘	28.687	6.756	4.246	<0.001	1.482	0.675
‘oil_heattype‘	6960.099	3195.621	2.178	0.030	1.063	0.941
‘electric_heattype‘	−1796.129	2739.932	−0.656	0.512	1.079	0.927
‘baths‘	6859.544	1985.183	3.455	<0.001	2.252	0.444
‘bedrooms‘	3800.843	1585.414	2.397	0.017	1.414	0.707
‘fireplcs‘	2994.806	1153.826	2.596	0.010	1.642	0.609
‘baseste‘1	−3277.464	3284.869	−0.998	0.319	1.220	0.820
‘crwlarea‘	−18.699	9.629	−1.942	0.053	10.823	0.092
‘bsmtarea‘	−8.584	9.162	−0.937	0.349	19.157	0.052
‘conduit‘1	−1358.003	2388.589	−0.569	0.570	3.028	0.330

Note: Coefficient values populated from Model 3 output. Tolerance = 1/VIF.

Table A46: Model 4 Coefficients and Collinearity Statistics

Variable	Estimate	Std. Error	t value	Pr(>|t|)	VIF	Tolerance
(Intercept)	−2108.302	7700.437	−0.274	0.784	—	—
‘adjusted_land_size‘	9.460	0.748	12.648	<0.001	1.737	0.576
‘dlin_flrarea1‘	81.665	8.622	9.472	<0.001	8.198	0.122
‘dlin_flrarea2‘	47.393	7.655	6.191	<0.001	1.496	0.669
‘dlin_bsmtfin‘	19.614	4.005	4.897	<0.001	1.953	0.512
‘dlin_gar_area_attached‘	53.424	8.856	6.032	<0.001	1.380	0.725
‘dlin_gar_area_detached‘	29.551	8.133	3.633	<0.001	1.202	0.832
‘dlin_gar_area_basement‘	17.009	3.393	5.014	<0.001	1.643	0.609
‘dlin_cp_area‘	29.192	6.755	4.321	<0.001	1.485	0.673
‘oil_heattype‘	6825.941	3173.047	2.151	0.032	1.051	0.952
‘electric_heattype‘	−1638.181	2728.080	−0.600	0.548	1.072	0.932
‘baths‘	6576.630	1992.411	3.301	0.001	2.274	0.440
‘bedrooms‘	3926.789	1585.571	2.477	0.014	1.417	0.705
‘fireplcs‘	3127.965	1158.834	2.699	0.007	1.661	0.602
‘baseste‘1	−3537.982	3281.881	−1.078	0.282	1.221	0.819
‘crwlarea‘	−20.534	9.600	−2.139	0.033	10.784	0.093
‘bsmtarea‘	−10.222	9.164	−1.115	0.265	19.214	0.052
‘alley‘1	2476.239	2035.853	1.216	0.225	2.084	0.480

Note: Coefficient values populated from Model 4 output. Tolerance = 1/VIF.

E.3 Stepwise Regression

The following tables provide detailed coefficient estimates and collinearity statistics (Variance Inflation Factor - VIF, Tolerance) for each step of the stepwise MRA model calibration performed on the MODEL dataset.

Table A47: Stepwise Regression - Step 1 Coefficients and Collinearity Statistics

Variable	Estimate	Std. Error	t value	Pr(>|t|)	VIF	Tolerance
(Intercept)	−9833.714	8376.466	−1.174	0.241	—	—
adjusted_land_size	10.247	0.887	11.549	<0.001	1.653	0.605
dlin_flrarea1	75.525	6.017	12.552	<0.001	2.444	0.409
dlin_flrarea2	64.694	11.244	5.753	<0.001	1.405	0.712
dlin_bsmtfin	22.933	4.845	4.733	<0.001	1.732	0.577
dlin_gar_area_attached	40.717	11.459	3.553	<0.001	1.347	0.742
dlin_gar_area_detached	29.308	12.254	2.392	0.017	1.161	0.861
dlin_gar_area_basement	13.625	4.302	3.167	0.002	1.895	0.528
dlin_cp_area	23.825	8.111	2.937	0.004	1.432	0.698
oil_heattype	2179.804	4383.208	0.497	0.619	1.089	0.918
electric_heattype	462.383	3273.741	0.141	0.888	1.077	0.929
baths	4853.821	2510.232	1.934	0.054	2.163	0.462
bedrooms	3320.709	1912.951	1.736	0.084	1.268	0.788
fireplcs	2399.663	1389.194	1.727	0.085	1.527	0.655
baseste1	−4921.182	4347.443	−1.132	0.259	1.157	0.864
crwlarea	−13.624	4.269	−3.192	0.002	1.215	0.823
alley1	4794.670	2416.183	1.984	0.048	1.821	0.549

Note: Tolerance = 1/VIF.

Table A48: Stepwise Regression - Step 2 Coefficients and Collinearity Statistics

Variable	Estimate	Std. Error	t value	Pr(>|t|)	VIF	Tolerance
(Intercept)	−9785.752	8355.619	−1.171	0.242	—	—
adjusted_land_size	10.249	0.886	11.572	<0.001	1.652	0.605
dlin_flrarea1	75.494	6.003	12.576	<0.001	2.441	0.410
dlin_flrarea2	64.710	11.225	5.765	<0.001	1.405	0.712
dlin_bsmtfin	22.952	4.835	4.747	<0.001	1.731	0.578
dlin_gar_area_attached	40.936	11.335	3.611	<0.001	1.323	0.756
dlin_gar_area_detached	29.285	12.232	2.394	0.017	1.161	0.862
dlin_gar_area_basement	13.646	4.293	3.179	0.002	1.893	0.528
dlin_cp_area	23.925	8.066	2.966	0.003	1.421	0.704
oil_heattype	2171.715	4375.523	0.496	0.620	1.089	0.918
baths	4838.371	2503.664	1.933	0.054	2.159	0.463
bedrooms	3314.639	1909.278	1.736	0.084	1.268	0.789
fireplcs	2415.913	1382.112	1.748	0.082	1.517	0.659
baseste1	−4839.412	4301.534	−1.125	0.261	1.137	0.880
crwlarea	−13.590	4.254	−3.194	0.002	1.211	0.826
alley1	4766.854	2404.127	1.983	0.048	1.809	0.553

Note: Tolerance = 1/VIF. Variable ‘electric_heattype‘ removed.

Table A49: Stepwise Regression - Step 3 Coefficients and Collinearity Statistics

Variable	Estimate	Std. Error	t value	Pr(>|t|)	VIF	Tolerance
(Intercept)	−9717.292	8343.801	−1.165	0.245	—	—
adjusted_land_size	10.272	0.883	11.628	<0.001	1.648	0.607
dlin_flrarea1	75.420	5.994	12.584	<0.001	2.439	0.410
dlin_flrarea2	64.486	11.202	5.757	<0.001	1.403	0.713
dlin_bsmtfin	22.901	4.828	4.743	<0.001	1.730	0.578
dlin_gar_area_attached	40.758	11.315	3.602	<0.001	1.322	0.757
dlin_gar_area_detached	30.315	12.039	2.518	0.012	1.127	0.887
dlin_gar_area_basement	13.775	4.279	3.219	0.001	1.886	0.530
dlin_cp_area	23.931	8.056	2.971	0.003	1.421	0.704
baths	4919.547	2495.123	1.972	0.050	2.149	0.465
bedrooms	3240.325	1900.965	1.705	0.089	1.260	0.794
fireplcs	2371.231	1377.414	1.722	0.086	1.510	0.662
baseste1	−4956.315	4289.590	−1.155	0.249	1.133	0.883
crwlarea	−13.803	4.227	−3.265	0.001	1.198	0.835
alley1	4897.168	2386.693	2.052	0.041	1.787	0.560

Note: Tolerance = 1/VIF. Variables ‘electric_heattype‘, ‘oil_heattype‘ removed.

Table A50: Stepwise Regression - Step 4 Coefficients and Collinearity Statistics (Final Model)

Variable	Estimate	Std. Error	t value	Pr(>|t|)	VIF	Tolerance
(Intercept)	−10 790.0	8296.0	−1.301	0.194	—	—
adjusted_land_size	10.370	0.880	11.782	<0.001	1.633	0.612
dlin_flrarea1	74.560	5.950	12.530	<0.001	2.401	0.416
dlin_flrarea2	64.280	11.210	5.736	<0.001	1.402	0.713
dlin_bsmtfin	21.640	4.707	4.599	<0.001	1.642	0.609
dlin_gar_area_attached	41.090	11.320	3.630	<0.001	1.321	0.757
dlin_gar_area_detached	29.250	12.010	2.435	0.015	1.121	0.892
dlin_gar_area_basement	13.730	4.282	3.207	0.001	1.886	0.530
dlin_cp_area	23.640	8.057	2.934	0.004	1.420	0.704
baths	5645.000	2416.000	2.336	0.020	2.013	0.497
bedrooms	3290.000	1902.000	1.730	0.085	1.259	0.794
fireplcs	2346.000	1378.000	1.703	0.090	1.510	0.662
crwlarea	−13.610	4.226	−3.220	0.001	1.196	0.836
alley1	4726.000	2383.000	1.983	0.048	1.780	0.562

Note: Tolerance = 1/VIF. Variable ‘baseste‘ removed.

E.4 Model Testing

The following tables present detailed results from testing the calibrated MRA model on the MODEL dataset (N=310).

Table A51: Kruskal-Wallis Test: Final MRA ASR by Neighbourhood (MODEL Set - After Adjustment)

Statistic	Value
Kruskal-Wallis chi-squared	9.654
df	7
p-value	0.209

Table A52: Ratio Statistics for Final MRA ASR by Neighbourhood (MODEL Set - After Adjustment)

Neigh.	Median	Mean	LB Mean	UB Mean	Min	Max	COD (%)	CV (%)	LB Median	UB Median
713	1.01	0.97	0.95	0.99	0.77	1.33	8.33	11.30	0.99	1.03
715	1.01	1.02	0.99	1.05	0.79	1.30	7.05	9.32	0.97	1.04
716	1.01	1.00	0.99	1.01	0.83	1.17	5.97	7.53	0.99	1.02
731	0.95	0.96	0.92	0.99	0.85	1.07	5.83	7.39	0.92	1.01
732	1.03	1.00	0.89	1.10	0.85	1.06	5.19	8.66	NA	NA
734	1.00	0.99	0.91	1.06	0.75	1.12	7.14	10.58	0.91	1.03
736	0.98	0.97	0.93	1.01	0.80	1.12	6.55	8.22	0.91	1.01
737	1.00	1.01	0.98	1.04	0.82	1.29	8.07	10.25	0.96	1.03

Table A53: Kruskal-Wallis Test Results: Final MRA ASR by Other Variables (MODEL Set)

Variable Tested	p-value
baseste	0.378
heattype	0.152

Table A54: Ratio Statistics for MRA ASR by Neighbourhood (FULL Dataset - Before Final Adjustment)

Neigh.	Median	Mean	LB Mean	UB Mean	Min	Max	COD (%)	CV (%)	LB Median	UB Median
713	1.02	1.04	1.02	1.07	0.77	1.34	8.67	11.25	1.01	1.04
715	1.00	1.01	0.98	1.03	0.79	1.30	7.00	9.04	0.97	1.03
716	1.00	1.00	0.99	1.01	0.83	1.18	6.22	7.69	0.99	1.02
731	0.95	0.96	0.92	1.00	0.79	1.27	7.79	10.63	0.91	0.98
732	1.01	1.00	0.92	1.08	0.85	1.14	6.85	9.49	0.85	1.06
734	1.00	0.99	0.92	1.05	0.75	1.12	7.71	10.61	0.91	1.03
736	0.98	0.97	0.94	1.00	0.80	1.12	6.15	7.58	0.92	1.01
737	1.01	1.01	0.99	1.03	0.82	1.29	7.48	9.55	0.97	1.03

Table A55: Kruskal-Wallis Test: Final MRA ASR by Neighbourhood (FULL Set - After Adjustment)

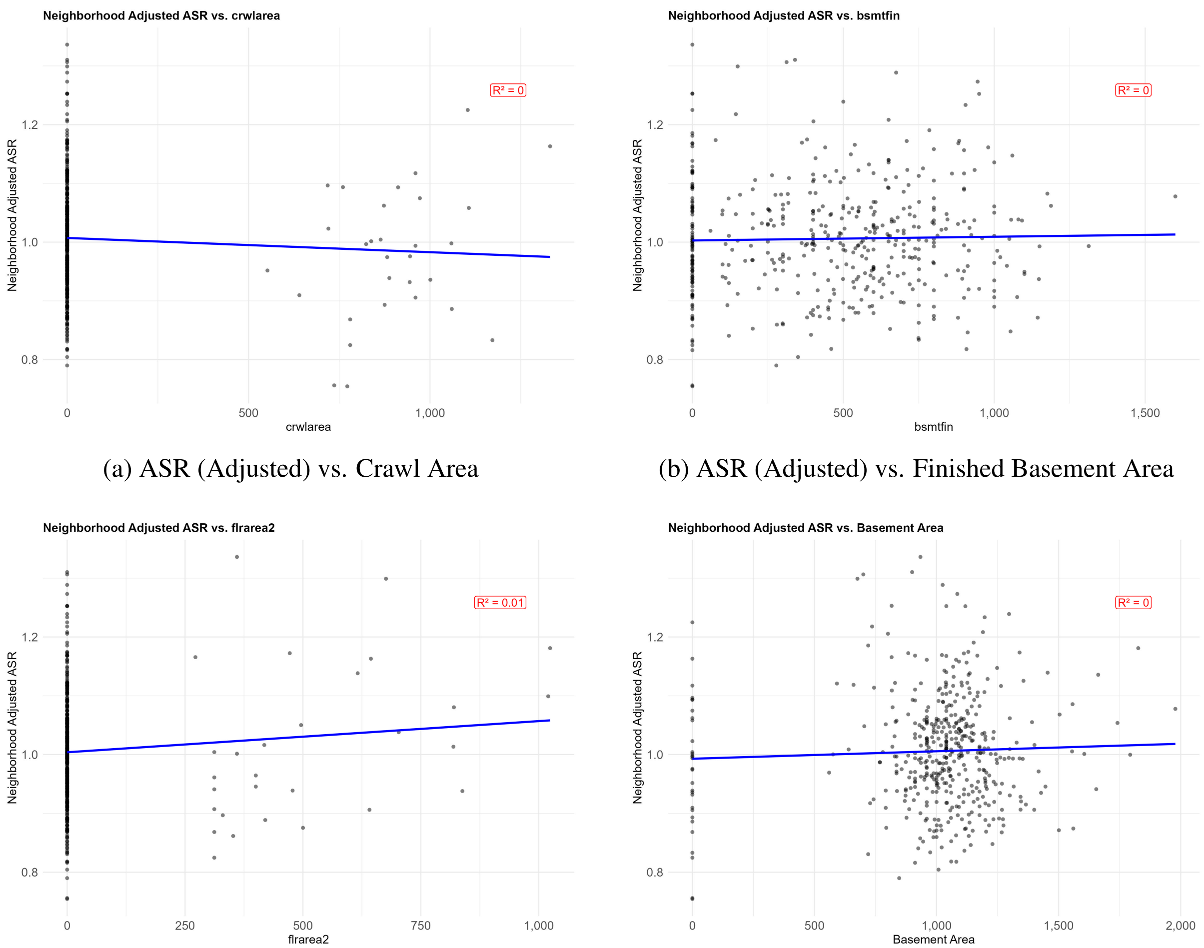
Statistic	Value
Kruskal-Wallis chi-squared	6.059
df	7
p-value	0.533

Table A56: Ratio Statistics for Final MRA ASR by Neighbourhood (FULL Set - After Adjustment)

Neigh.	Median	Mean	LB Mean	UB Mean	Min	Max	COD (%)	CV (%)	LB Median	UB Median
713	1.00	1.02	1.00	1.05	0.76	1.31	8.67	11.25	0.99	1.02
715	1.00	1.01	0.98	1.03	0.79	1.30	7.00	9.04	0.97	1.03
716	1.00	1.00	0.99	1.01	0.83	1.18	6.22	7.69	0.99	1.02
731	1.00	1.01	0.97	1.05	0.83	1.34	7.79	10.63	0.96	1.03
732	1.01	1.00	0.92	1.08	0.85	1.14	6.85	9.49	0.85	1.06
734	1.00	0.99	0.92	1.05	0.75	1.12	7.71	10.61	0.91	1.03
736	0.98	0.97	0.94	1.00	0.80	1.12	6.15	7.58	0.92	1.01
737	1.01	1.01	0.99	1.03	0.82	1.29	7.48	9.55	0.97	1.03

Table A57: Kruskal-Wallis Test Results: Final MRA ASR by Other Variables (FULL Set)

Variable Tested	p-value
heattype	0.157
gar_type	0.771
conduit	0.704
baths	0.824
bedrooms	0.100
fireplcs	0.620
baseste	0.389
carptfl	0.310
gr_mancl	0.132
cp_mancl	0.469



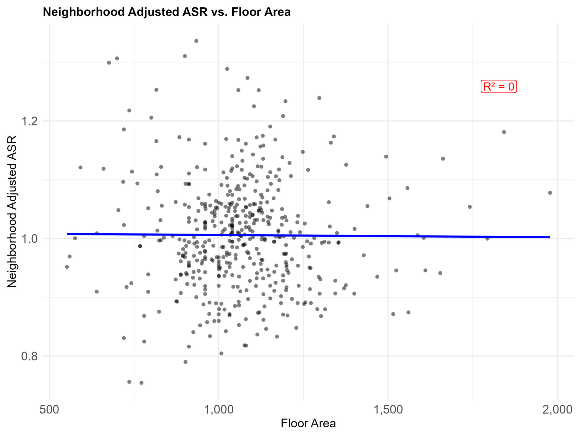
(a) ASR (Adjusted) vs. Crawl Area

(b) ASR (Adjusted) vs. Finished Basement Area

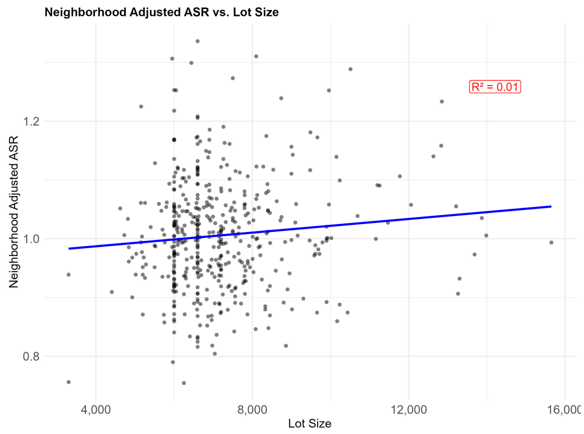
(c) ASR (Adjusted) vs. Floor Area 2

(d) ASR (Adjusted) vs. Basement Area

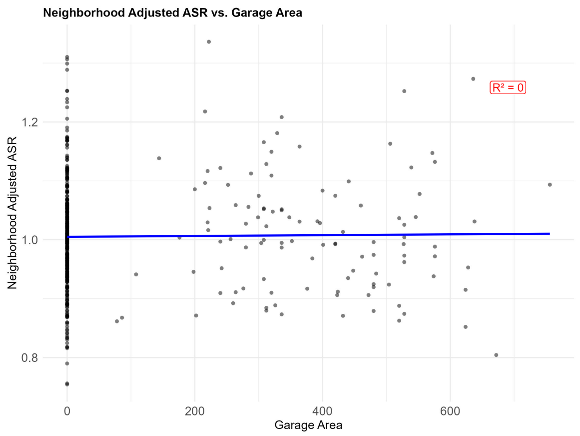
Figure A1: Scatter Plots of Assessed Sale Ratio (Neighborhood 2) vs. Various Area Variables



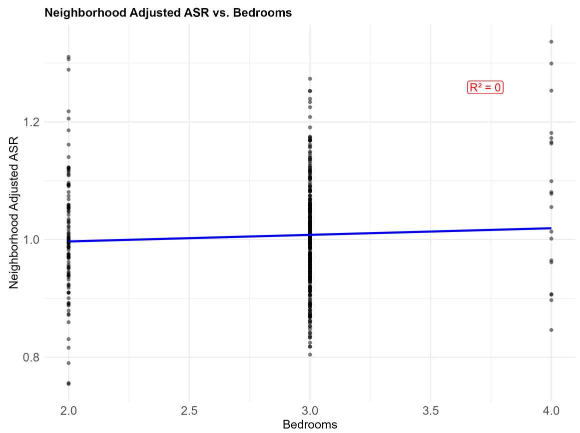
(a) ASR (Adjusted) vs. Floor Area 1



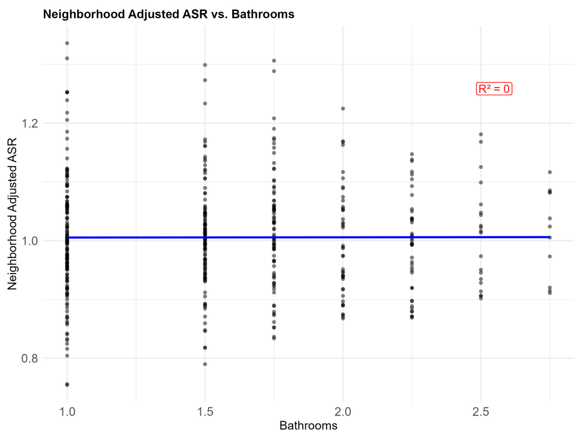
(b) ASR (Adjusted) vs. Lot Size



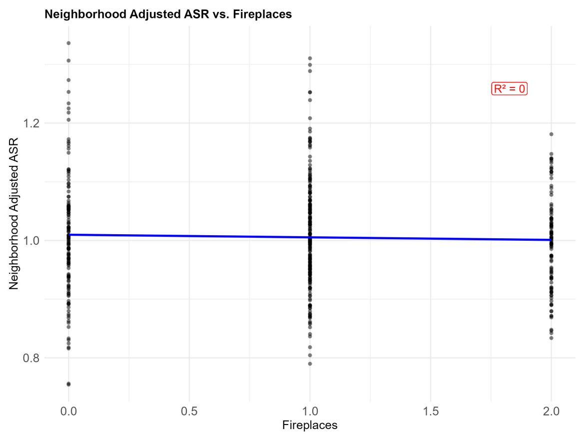
(c) ASR (Adjusted) vs. Garage Area



(d) ASR (Adjusted) vs. Bedrooms



(e) ASR (Adjusted) vs. Baths



(f) ASR (Neighborhood Adjusted) vs. Fireplaces

Figure A2: Scatter Plots of Assessed Sale Ratio (Neighborhood 2) vs. Various Property Characteristics

F Syntax

Full R syntax is attached separately; this section only focuses on the variable transformation that is referred in the main text.

F.1 Transformation

```
### Variable Transofrmation ###
sales_data%<>%mutate(adjusted_land_size = land_value_final
/ median_size_factor,
 dlin_flrarea1 = MB_factor * flrarea1 * pctgood,
 dlin_flrarea2 = MB_factor * flrarea2 * pctgood,
 dlin_bsmtfin = MB_factor * bsmtfin * pctgood,
 dlin_gar_area_attached = CP_factor * gar_area * (gar_type == 1) * pctgood,
 dlin_gar_area_detached = CP_factor * gar_area * (gar_type == 2) * pctgood,
 dlin_gar_area_detached = CP_factor * flrarea1 * (gar_type == 3) * pctgood,
 dlin_cp_area = CP_factor * cp_area * pctgood,
 gas_heattype = ifelse(heattype %in% c(1,2,7),1,0),
 oil_heattype = ifelse(heattype %in% c(3,8),1,0),
 electric_heattype = ifelse(heattype %in% c(4,5,6),1,0))
```